

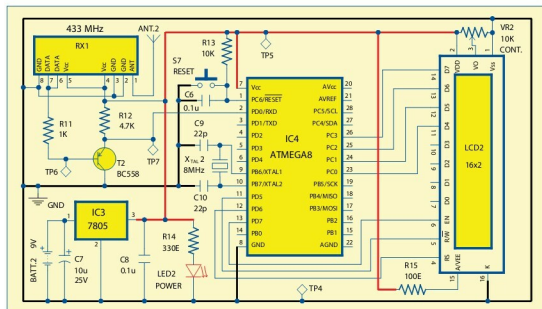


Fig. 4: Circuit diagram of the receiver section

PARTS LIST

Semiconductors:

- IC1, IC3 - 7805, 5V regulator
 IC2, IC4 - ATmega8, CMOS 8-bit microcontroller
 T1, T2 - BC558 npn transistor
 LCD1, LCD2 - 16x2 LCD display
 LED1, LED2 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R12 - 4.7-kilo-ohm
 R2, R11 - 1-kilo-ohm
 R3, R14 - 330-ohm
 R4, R15 - 100-ohm
 R5-R9, R10, R13 - 10-kilo-ohm
 VR1, VR2 - 10-kilo-ohm preset

Capacitors:

- C1, C3, C6, C8 - 0.1 μ F ceramic
 C2, C7 - 10 μ F, 25V electrolytic
 C4-C5, C9, C10 - 22pF ceramic

Miscellaneous:

- TX1 - 433MHz transmitter module
 RX1 - 433MHz receiver module
 BATT.1, BATT.2 - 9V battery
 S1-S7 - Tactile switch
 X_TAL.1, X_TAL.2 - 8MHz crystal oscillator
 ANT.1, ANT.2 - 17cm single-wire antenna

Keypad interface. The keypad consists of five tactile switches (S2 through S6), where each switch is assigned a particular function for up, down, enter, clear and bill functions in the transmitter unit. There is one reset switch each in transmitter unit and receiver unit, which is used if/when there is a problem in the LCD display.

LCD display module. The 16x2 LCD display is used for the user interface. The LCD is directly controlled

by the MCU. The LCD used is a 16x2 (16 characters by two lines) format type, driven by Hitachi HD44780 or an equivalent driver controller. The LCD module can display all alphabets and symbols as per ASCII format, and can store about eight custom characters.

The HD44780 has two 8-bit registers, an instruction register (IR) and a data register (DR). The IR stores instruction codes such as display clear and cursor shift, and address information for display data RAM (DD RAM) and character generator RAM (CG RAM). The IR can be written from the MCU but not read by it.

The DR temporarily stores data to be written into DD RAM or CG RAM and data to be read out from DD RAM or CG RAM. Data written into the DR from the MCU is automatically written into DD RAM or CG RAM by internal operation.

The DR is also used for data storage when reading from DD RAM or CG RAM. When address information is written into the IR, data is read into the DR from DD RAM or CG RAM by internal operation. Data transfer to the MCU is then completed by the MCU reading the DR.

RF transmitter module. The RF433MHz transmitter module is for

ASK modulation using an RF transmitter tuned to 433MHz frequency. It features an in-built crystal oscillator and can be used to send bit-wise transmission. Orders given by the customer are encoded and sent wirelessly through this transmitter. The power supply can vary from 4V to 12V, depending on the range and power required. Data pin of the transmitter module is connected to TXD pin of IC2

through transistor T1 and resistor R2. An antenna of 15cm to 20 cm length is desirable for long-range communication.

RF receiver module. The RF433MHz receiver module can work in sync with RF433MHz transmitter. It receives signals sent by the transmitter. Receiver data pins are connected to RXD pin of IC4 through transistor T2. The receiver only needs 5V supply and an external antenna. This module is ideal for short-range remote-controlled applications, where cost is the primary concern.

Power supply. The system requires a regulated voltage of +5V for the MCU, LCD and RF modules. The battery voltage is converted to 5V, required for both transmitter and receiver units using IC 7805 regulator.

Capacitors are used in the input and output of IC 7805 to have more stability and to bypass transient signals. An LED in series with a current-limiting resistor acts as a power-on indicator.

Clock inputs. Crystal oscillators X_TAL.1 and X_TAL.2 are used as external clock sources in this project. Either a quartz crystal or a ceramic resonator may be used. For resonators, the maximum frequency is 8MHz if CKOPT fuse bit is unprogrammed, and 16MHz